

Optimized 4. Constrained Optimization

Agenda

- Recap
- coding gradient descent
- computing gradients of fnet
- optimization problem
- Constrained optimization problem
 - * How to convert constraint optimization problem into unconstrained optimization problem
 - * Generalizing constraint optimization problem
 - * Example of solving a constraint optimization problem
 - * solving original constraint optimization problem
 - computing gradient of loss fnet with respect to w vector
 - computing gradient of loss fnet w.r.t. w_0

partial derivative code

```
def get-partial (func, x, y)
    delta = 0.0000001
    fx = (func(x+delta, y) - func(x, y)) / delta
    fy = (func(x, y+delta) - func(x, y)) / delta
    return fx, fy
```

```
def func(x, y)
    return (x**2 + y**2)
```

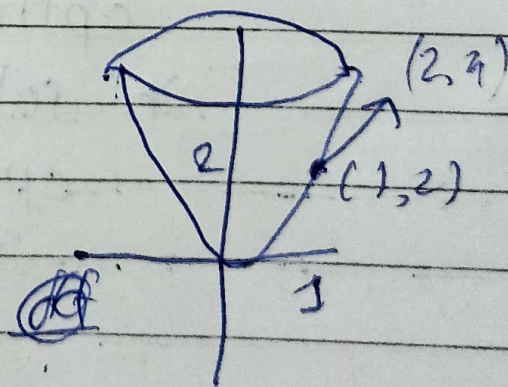
$$f = x^2 + y^2$$

$$\frac{df}{dx} = 2x \quad \frac{df}{dy} = 2y$$

$$\text{gradient} = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

$$\text{at } (x, y) = (1, 2)$$

$$\text{gradient} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$



we get same by calling `finet`.

`get-partial (func, 1, 2)`

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look at the colab notebook to
look at code.

Optimization problem

We have dataset find $\bar{w}, w_0$
to have maximum gain `finet`

$$G(D, \bar{w}, w_0) = \sum_{i=1}^N \frac{w^T x_i + w_0}{\|w\|} y_i$$

maximize $G(D, \bar{w}, w_0)$

$$L(D, \bar{w}, w_0) = -G(D, \bar{w}, w_0)$$

minimize $L(D, \bar{w}, w_0)$

$$= - \sum \left(\frac{\omega^T x_i + \omega_0}{\|\omega\|} \right) y_i$$

$$L = - \sum_i \left(\frac{\omega_1 x_{i1} + \omega_2 x_{i2} + \dots + \omega_0}{\sqrt{\omega_1^2 + \omega_2^2 + \dots}} \right) y_i$$

Need to calculate $\frac{dL}{d\omega_1}$, $\frac{dL}{d\omega_2}$, ...

Then calculate new ω 's

$$\omega_1 = \omega_1^{\text{old}} - \eta \frac{dL}{d\omega_1}$$

$$\omega_2 = \omega_2^{\text{old}} - \eta \frac{dL}{d\omega_2}$$

:

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Here loss function is in quotient format i.e. $f(x)/g(x)$ format

we need to do something like

$$\frac{g(x) f'(x) - f(x) g'(x)}{g(x)^2}$$

it will be complex, so try to get rid of $g(x)$

if $\sqrt{w_1^2 + w_2^2 + \dots} = 1$
then problem solved.

then

$$L = \sum w_i$$

This is called constrained problem

constraint is $\|w\| = 1$

↓
condition imposed

$$\text{minimize} \quad -\sum_i \frac{\omega^T x_i + \omega_0}{\|\omega\|} y_i$$

$$\text{where } \|\omega\| = 1$$

It is constrained optimization problem

e.g. $3x + 4y + 4 = 0$
 $\omega_1 x + \omega_2 y + \omega_0 = 0$

$$\|\omega\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

New line will be

$$\frac{3}{5}x + \frac{4}{5}y + \frac{4}{5} = 0$$

$$\omega_1 x + \omega_2 y + \omega_0 = 0$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ \frac{\omega_1}{\|\omega\|} & \frac{\omega_2}{\|\omega\|} & \frac{\omega_0}{\|\omega\|} \end{array}$$

Problem with constraint problem

Here let's say

$$y = w_2 x_2 + w_1 x_1 + w_0$$

Initially we can select random w_2 , w_1 , w_0 such that $\|w\| = 1$

we update

$$w_{2, \text{new}} = w_2 - \eta \frac{d}{dw_2} y \quad \text{similarly we get } w_1, w_0$$

but these it no guarantee

new w' vectors will be unit vector.

② Why can't we divide new w' & get unit w'_{unit}

$$\text{like a } w'_{\text{unit}} = \frac{w'}{\|w'\|} \quad ?$$

So this is not useful.

because of this new concept is there
called Lagrange Multiplier

if you have a function $f(x)$ &
 $c(x)$ is constraint then you can
get new function such that
 $f(x) + \lambda (c(x))$

~~if you~~

Then minimum at new function
will be minimum at $f(x)$

$$\text{Here } f(x) = \min \sum \frac{\omega^T x_i}{\|\omega\|} y_i$$

$$c(x) = \|\omega\| - 1 = 0$$

New function

$$\min \sum \frac{\omega^T x_i}{\|\omega\|} y_i + \lambda (\|\omega\| - 1)$$

Example Lagrange multiplier

we want to find min of

$$f = x^2 - 3x + 3$$

that is optimization problem

we want to find min of

$$f = x^2 - 3x + 3$$

such that $-x^2 + 2x + 3 = 0$

this is constrained optimization problem, we can use Lagrange here.

New function

$$x^2 - 3x + 3 + \lambda (-x^2 + 2x + 3)$$

Need

$$f = x^2 - 3x - 3 - \lambda x^2 + 2x\lambda + 3\lambda$$

$$\frac{df}{dx} = 2x - 3 - 2\lambda x + 2\lambda$$

$$= (\cancel{2 - 2\lambda})x - 1 \quad \lambda = \frac{2x - 3}{2(x - 1)}$$

min value will be at ~~$x = \frac{2x - 3}{2(x - 1)}$~~

$$\frac{df}{d\lambda} = -x^2 + 2x + 3 = 0 \Rightarrow x = 3, -1$$

min value will be at $x = 3, -1$

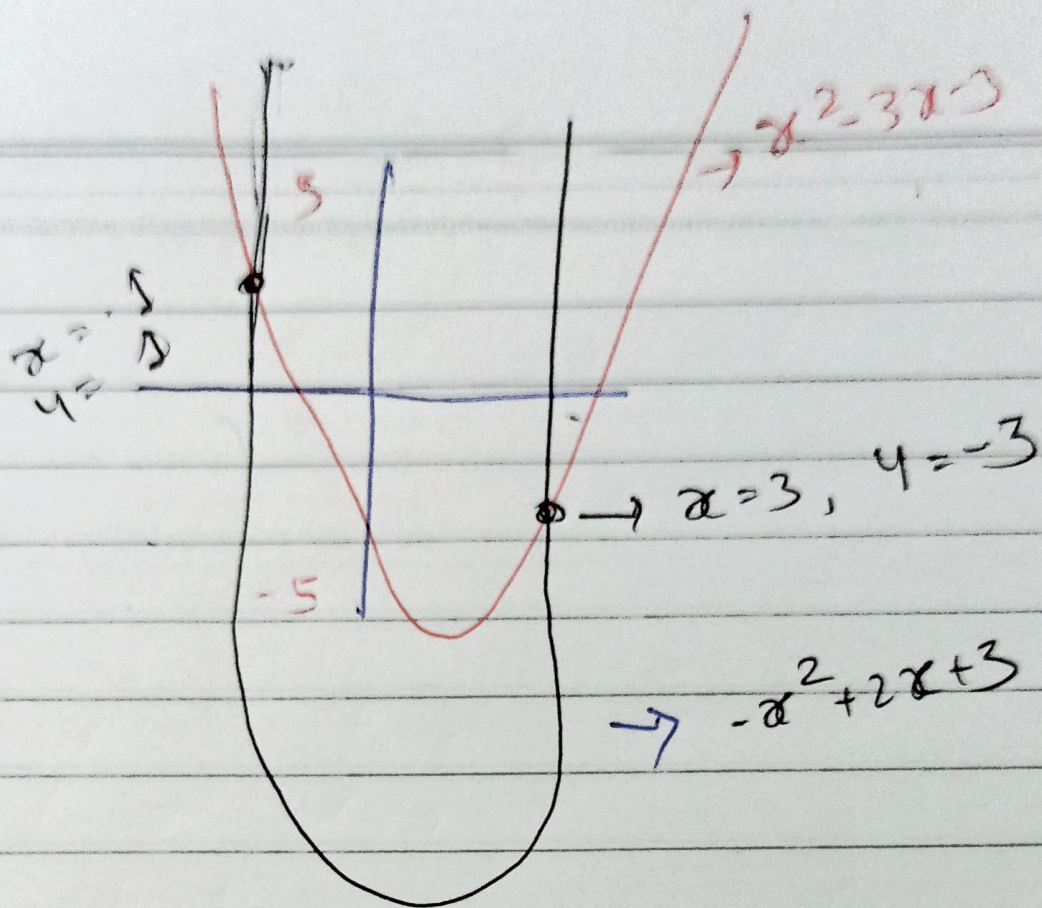
$$\begin{array}{ll} \text{at } x = 3 & \lambda = 3/4 \\ x = -1 & \lambda = 5/4 \end{array}$$

so min value will be

$$x = 3 \quad \lambda = 3/4 \quad \text{min value } -3$$

$$x = -1 \quad \lambda = 5/4 \quad \text{min value } -1$$

Graph



look notes for longrange multiplier
for loss function
